

oncogene sequence into the genome of the non-human mammal did not involve an "essentially biological process". The product claim for the genetically-manipulated animal included descendants not directly genetically manipulated themselves but produced by the essentially biological process of sexual reproduction. The board held that this was a product claim defined in terms of the process by which it was produced and that a product-by-process claim remained a product claim irrespective of the process it referred to.

In T.315/03 (OJ 2006, 15) the board held that the claimed process for producing transgenic mice, which included genetic manipulation, was not "an essentially biological process", the latter being defined in R.23b(5) EPC 1973 (R.26(5) EPC) as consisting "entirely of natural phenomena such as crossing or selection".

In T.1199/08 appellant II argued that if the technical teaching of the patent as a whole was taken into account, it became obvious that the only purpose of the claimed method was the production of an animal. The board rejected the argument that the "whole contents approach" developed in G.2/06 should be taken into consideration for a correct assessment of whether the claim in question would fall under the prohibition of Art. 53(b) EPC. It also decided that a claim to a method which allowed for the long-term preservation of sex-selected sperm did not qualify as an essentially biological process for the production of animals. Therefore, in line with the parallel reasoning developed in G.2/07 regarding processes for the production of plants (see below), it did not fall under the prohibition of Art. 53(b) EPC.

3.3.2 Essentially biological processes for the production of plants

a) G 2/07 and G 1/08

The leading cases concerning the interpretation of the exclusion from patentability of essentially biological processes for the production of plants are the consolidated cases G.2/07 (OJ 2012, 130) and G.1/08 (OJ 2012, 206) ruling on the referral in T.83/05 of 22 May 2007 (OJ 2007, 644) relating to a method of obtaining particular broccoli lines and on that in T.1242/06 of 4 April 2008 (OJ 2008, 523) relating to a method of breeding tomato plants which produce tomatoes with reduced fruit water content.

The Enlarged Board of Appeal found, first of all, that the exception applying to "essentially biological processes for the production of plants" could not be read as limited to processes for the production of plant varieties, because such an interpretation would be contrary to the wording.

After rejecting a series of possible approaches to the interpretation of the exclusion, the Enlarged Board concluded that a non-microbiological process for the production of plants which contains or consists of the steps of sexually crossing the whole genomes and of subsequently selecting plants is "essentially biological" within the meaning of Art. 53(b) EPC. Such a process does not escape the exception to patentability merely because it contains, as a further step or as part of any of the steps of crossing and selection, a step of a technical nature which serves to enable or assist performance of the

steps of sexually crossing the whole genomes of plants or of subsequently selecting plants.

This outcome was largely based on the Enlarged Board's findings with regard to the legislative history of the Strasbourg Patent Convention and the EPC 1973. The legislator's intention, it found, had been to exclude from patentability those plant breeding processes which were the conventional methods of plant-variety breeding at the time. These conventional methods included, in particular, those based on the sexual crossing of plants (i.e. of their whole genomes) deemed suitable for the purpose pursued and on the subsequent selection of the plants having the desired trait(s). It could also be gathered from the legislative history that the mere use of a technical device in a breeding process was not to be considered sufficient to lend the process itself a technical character.

The Enlarged Board distinguished such processes from those which left the realm of plant breeding. It cited R. 27(c) EPC, which expressly provides that biotechnological inventions are also patentable if they concern a microbiological or other technical process, so that the excluded essentially biological processes are juxtaposed with the patentable technical processes. The exception to patentability under Art. 53(b) EPC does not exclude a process of sexual crossing and selection which includes within it an additional step of a technical nature, which step by itself introduces a trait into the genome or modifies a trait in the genome of the plant produced, so that the introduction or modification of that trait is not the result of the mixing of the genes of the plants chosen for sexual crossing. In the context of examining whether such a process is excluded from patentability as being "essentially biological" within the meaning of Art. 53(b) EPC, it is not relevant whether a step of a technical nature is a new or known measure, whether it is trivial or a fundamental alteration of a known process, whether it does or could occur in nature or whether the essence of the invention lies in it.

Ultimately, this means that, while the presence in a claim of one feature which could be characterised as biological does not necessarily result in exclusion of the claimed process as a whole under Art. 53(b) EPC, the same does not apply where the process includes sexual crossing and selection.

In T. 2323/11 the invention concerned the removal of unwanted sequences from the genome of transgenic plants which comprise an introduced gene for an agronomically valuable trait. In G 2/07 and G 1/08 the Enlarged Board considered that, under Art. 53(b) EPC, excluded processes were characterised by the fact that the traits of the plants resulting from the crossing were determined by the underlying natural phenomenon of meiosis. The board held – contrary to the appellant's argument – that the trait of the excision of the target gene is the result of the crossing of the parent plants and is determined by the underlying natural phenomenon of meiosis, as the latter determines the genetic make-up of the plants produced. The claimed method therefore did not fall under the exception made in G 2/07 and G 1/08 for those processes which contain an additional step of technical nature.

In T. 2435/13 the board dealt with the interpretation of the term "essentially biological process for the production of plants" in the light of G 2/07 and G 1/08. The claimed subject-

matter was directed to obtaining a Brassica oleracea plant resistant to clubroot by crossing a Brassica rapa plant resistant to clubroot with a B. oleracea plant, rescuing embryos resulting from the cross and regenerating plants from them, selecting plants that were resistant to clubroot, back-crossing such plants with a B. oleracea plant, rescuing embryos resulting from the back-cross, and again selecting a plant that was resistant to clubroot. The board held that all methods containing or consisting of steps of sexually crossing the whole genome of plants and the subsequent selection of plants, regardless of those plants being sexually incompatible or not, to be excluded from patentability and not just those methods involving meiotic recombination events throughout the whole genome during perfect pairing of homologous chromosomes in the process of meiosis. Furthermore, the board held that while the embryo rescue steps were additional steps of a technical nature, as defined in answer 3 of decisions G 2/07 and G 1/08, the embryo rescue steps by themselves did not introduce the clubroot resistance trait into the genome of the plants produced. The claimed subject matter was held to be excluded from patentability.

In T 915/10 the invention concerned soybean plants which had been genetically modified. The claimed method was defined solely by the technical process step of introducing a gene sequence into the genome of the plant by a genetic engineering step introducing heterologous DNA in plant cells. The board noted that the introduced trait was due directly to the expression of the inserted DNA and was not the result of a plant breeding method characterised by crossing and selection. Indeed, the method as claimed did not require nor define steps of mixing genes of plants by sexual crossing and subsequent selection of plants, either explicitly or implicitly. The board was thus satisfied that the claimed method did not fall under the exclusion of "essentially biological processes for the production of plants" pursuant to Art. 53(b) EPC. The board considered that the claimed subject-matter was a method for the production of plants by means of genetic-engineering techniques, which involves laboratory techniques essentially different from breeding methods and which as such have been accepted in the case law to be patentable. The board noted that there was nothing in decisions G 2/07 and G 1/08 which would indicate that the Enlarged Board of Appeal was of the opinion that this practice ought to be reconsidered as a result of its analysis of the process exclusion in Art. 53(b) EPC.

b) Patentable technical processes

In T 1729/06 the invention was in the field of the production of watermelon fruit, in particular of seedless watermelons. The claimed use and methods concerned the pollination of the sterile female flowers of the triploid watermelon plant with pollen of the diploid polliniser plant. They did not concern sexually crossing two whole genomes of plants (implying meiosis and fertilisation) and the subsequent selection of plants. The board was therefore satisfied that the use and methods as subject-matter of the claims were not such methods which the Enlarged Board of Appeal in its decisions G 2/07 (OJ 2012, 130) and G 1/08 (OJ 2012, 206) had considered to fall under the exclusion of "essentially biological processes for the production of plants".

The board noted that, in those decisions, the Enlarged Board had not given a comprehensive and exhaustive definition of the subject-matter to which the process exclusion in Art. 53(b) EPC applies in relation to plant inventions. Therefore, it still needed

to be established whether the claimed uses and methods were excluded from patentability by virtue of the process exclusion in Art. 53(b) EPC for other reasons.

The board found that they were not. The legislator drafting Art. 53(b) EPC had not intended to exclude from patentability a whole class of inventions, i.e. horticultural or agricultural (agronomic) processes, under which the claimed use and methods undoubtedly fell. The EPC 1973 legislator (and the EPC 2000 legislator) had only wished to exclude from patentability the – then conventional – processes applied by plant breeders in connection with new plant varieties for which a special property right was available under the UPOV Convention, and processes which were fundamentally of this type.

Thus, rather than being excluded from patentability by virtue of Art. 53(b) EPC, the claimed uses and methods constituted a "technical process" and were therefore patentable pursuant to R. 27(c) EPC.

3.3.3 Product claims for plants or plant material

a) G 2/12 and G 2/13

In the consolidated cases G 2/12 and G 2/13 (OJ 2016, A28 and A29) the Enlarged Board was concerned with the question whether the exclusion of essentially biological processes for the production of plants in Art. 53(b) EPC had a negative effect on the allowability of product claims or product-by-process claims directed to plants or plant material (such as a fruit or plant parts) which are directly obtained and/or defined by an essentially biological process. It concluded that it did not.

The Enlarged Board applied the various methodical lines of interpretation set out in Art. 31 and 32 of the Vienna Convention on the Law of Treaties. None of these led the Enlarged Board to conclude that the term "essentially biological processes for the production of plants" extended beyond the processes to products defined or obtained by such processes. This result was confirmed when the preparatory work of the EPC was taken into account as a supplementary means of interpretation.

The Enlarged Board also examined whether considerations had arisen since the Convention was signed which might give reason to believe that a literal interpretation of the relevant provision would conflict with the legislator's aim. However, it could not see why the original intention of the legislator in drafting Art. 53(b) EPC was no longer justified, just because new plant breeding techniques were available today.

The Enlarged Board considered whether allowing patenting of a product claim or a product-by-process claim for a plant or plant material that is obtained by a means of an essentially biological process could be regarded as a circumvention of the process exclusion. It found that it could not, referring to the clear wording of Art. 53(b) EPC. It warned that broadening the scope of the process exclusion to the extent that it included products obtained by essentially biological processes for the production of plants would introduce an inconsistency in the system of the EPC, as plants and plant material other than plant varieties were generally eligible for patent protection.

On the question whether it was of relevance that the protection conferred by the product claim encompassed the generation of the claimed product by means of an essentially biological process for the production of plants excluded as such under Art. 53(b) EPC, the Enlarged Board highlighted the distinction between the requirements for patentability and the extent of protection. Whether a product claim or a product-by-process claim was patentable was to be examined irrespective of the extent of protection that is conferred by it after grant.

The Enlarged Board emphasised that its judicial decision-making powers did not extend to the ethical, social and economic aspects raised in the general debate, as it had no mandate to engage in legislative policy.

The questions of law referred to the Enlarged Board of Appeal were answered as follows:

1. The exclusion of essentially biological processes for the production of plants in Art. 53(b) EPC does not have a negative effect on the allowability of a product claim directed to plants or plant material such as a fruit (G 2/12) or plant parts (G 2/13).
2. The fact that the process features of a product-by-process claim directed to plants or plant material other than a plant variety define an essentially biological process for the production of plants does not render the claim unallowable (G 2/13). The fact that the only method available at the filing date for generating the claimed subject-matter is an essentially biological process for the production of plants disclosed in the patent application does not render a claim directed to plants or plant material other than a plant variety unallowable (G 2/12, G 2/13).
3. In the circumstances, it is of no relevance that the protection conferred by the product claim encompasses the generation of the claimed product by means of an essentially biological process for the production of plants excluded as such under Art. 53(b) EPC (G 2/12, G 2/13).

In its decision T 83/05 of 10 September 2015 the board applied G 2/13. Claim 1 of the main request was directed to an edible Brassica plant produced according to a method of crossing and selecting. Claims 2 and 3 were directed to an edible portion and to the seed of a broccoli plant produced by a method defined in the same manner as in claim 1. Claims 4 and 5 were directed to a broccoli plant and a broccoli inflorescence. The board remitted the case to the department of first instance with an order to maintain the patent on the basis of claims 1 to 5 of the main request.

In T 1242/06 of 8 December 2015 the board applied G 2/12. The newly submitted claims were restricted to claims for products and related to a (naturally) dehydrated tomato fruit of the species *L. esculentum*. The board held that the subject-matter in question was not excluded from patentability pursuant to Art. 53(b) EPC. The board remitted the case to the opposition division with the order to maintain the patent on the basis of these claims.

Following a notice of the European Commission, clarifying that Directive 98/44/EC of the European Parliament and of the Council of 6 July 1998 on the legal protection of